



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Rajarshi Majumder
14-Sep-26



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix



Executive Summary

- This is a Data Science project is to research and predict if SpaceX Falcon 9 will **land successfully**.
- SpaceX advertises Falcon 9 rocket launches on its website with a cost of **62M** USD primarily because of its booster's reusability capabilities.
- Through this research, if we can **predict** if the first stage will land, we can determine the cost of a launch.
- The findings from this research can be used by other Aerospace and Space Transport companies to be at par with Landing Success rates of SpaceX and at a competitive cost.



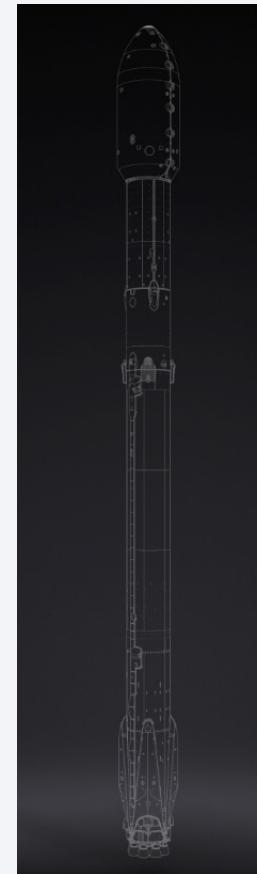
Introduction

Project background and context

- SpaceX is a leading space company dedicated to making space travel affordable and accessible. Its major achievements include delivering spacecraft to the International Space Station, deploying a satellite constellation that provides internet connectivity, and flying crewed missions.
- These accomplishments are enabled by relatively low launch costs of about \$62 M\$, made possible by the innovative reuse of the Falcon 9's first stage. Competitors that cannot recover and reuse the first stage typically charge \$165 M\$ or more per launch- ~ 166% higher.
- By predicting whether the first stage will successfully land, we can estimate the cost of a launch. Using publicly available data and machine learning models, we aim to forecast first-stage reuse success for SpaceX and other providers.

Project Success Criteria

- What key factors like payload mass, launch site, number of prior flights, and orbital placements influence first-stage landing success
- Trends in the rate of successful landings over a period of time
- The best-performing predictive model for successful landing (binary classification)



[Falcon 9](#) is a reusable, two-stage rocket designed and manufactured by SpaceX for the reliable and safe transport of people and payloads into Earth orbit and beyond. Falcon 9 is the world's first orbital class reusable rocket. Reusability allows SpaceX to re-fly the most expensive parts of the rocket, which in turn drives down the cost of space access.



Section 1

Methodology

Methodology

Executive Summary

1. Data collection:

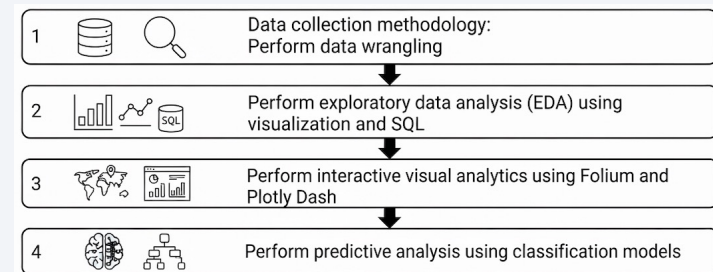
- SpaceX REST API for structured launch records
- Web scraping the Wikipedia “List of Falcon 9 and Falcon Heavy launches” page for supplementary launch data
- Data wrangling: cleaned missing values, filtered irrelevant orbit types, engineered a binary landing-success target

2. EDA: visualization (seaborn/matplotlib) and SQL queries (SQLite) to explore relationships between payload, orbit, site, and outcome

3. Interactive analytics: Folium map of launch sites and outcomes; Plotly Dash dashboard for launch-site and payload exploration

4. Predictive analysis: Trained Logistic Regression, SVM, Decision Tree, and KNN classifiers, tuned via GridSearchCV with 10-fold cross-validation

Methodologies Workflow Diagram

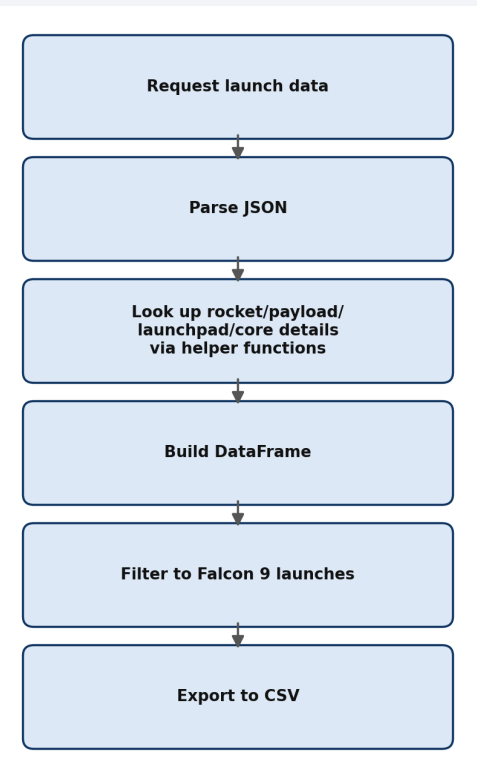


Data Collection

- Two complementary data sources were combined to build the full launch dataset:
 1. SpaceX REST API — queried `api.spacexdata.com/v4/launches/past` for structured records (rocket, payload, launchpad, core/landing outcome), then used helper functions to look up booster version, launch site, payload mass/orbit, and core/landing details for each launch
 2. Web scraping — parsed the Wikipedia Falcon 9/Falcon Heavy launches page with BeautifulSoup, extracting the launch table's rows into a structured DataFrame as a cross-check and supplement to the API data
- Both sources were merged and reconciled during data wrangling into a single clean dataset for analysis

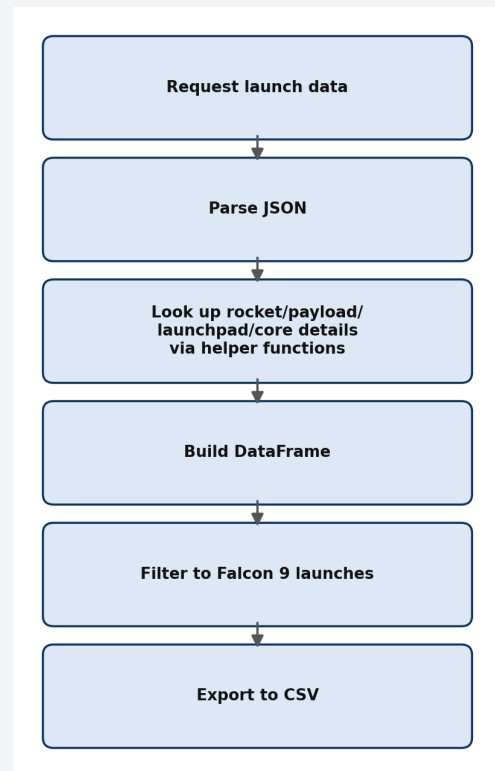
Data Collection – SpaceX API

- GET request to `api.spacexdata.com/v4/launches/past` returns all past launches as JSON
- Custom helper functions (`getBoosterVersion`, `getLaunchSite`, `getPayloadData`, `getCoreData`) enrich each launch record by following nested IDs (rocket, launchpad, payload, cores) back to the API for full details
- Results assembled into a pandas DataFrame, filtered to Falcon 9 only, and used as the primary structured data source
- GitHub URL: GitHub repo: <https://github.com/Raj-Majumder/IBM-Data-Science-Capstone-SpaceX-Falcon9-Landing->



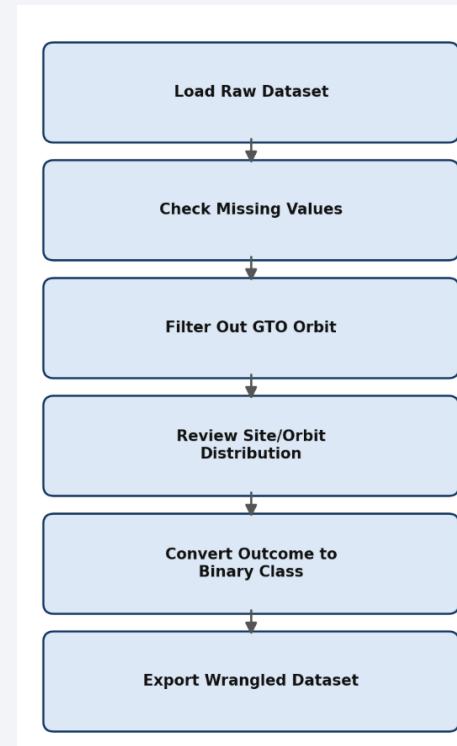
Data Collection - Scraping

- Requested the static Wikipedia revision of “List of Falcon 9 and Falcon Heavy launches” with a custom User-Agent header
- Parsed the HTML with BeautifulSoup, located the launch records table, and extracted column headers programmatically
- Iterated through table rows to build a launch_dict of columns (date/time, booster version, launch site, payload, orbit, outcome), then assembled into a DataFrame
- Exported the scraped result to spacex_web_scraped.csv
- GitHub URL: GitHub repo: <https://github.com/Raj-Majumder/IBM-Data-Science-Capstone-SpaceX-Falcon9-Landing->



Data Wrangling

- Checked for missing values across all columns and identified the LandingPad column as containing nulls where a launch had no landing attempt
- Removed launches to GTO orbit from the landing-outcome analysis, since geostationary transfer missions do not attempt first-stage recovery the same way
- Reviewed launch-site distribution (CCAFS SLC-40: 55, KSC LC-39A: 22, VAFB SLC-4E: 13) and orbit-type distribution across the remaining launches
- Converted the multi-category Outcome column (e.g. “True ASDS”, “False Ocean”) into a binary Class target: 1 for any successful landing, 0 otherwise
- GitHub URL: GitHub repo: <https://github.com/Raj-Majumder/IBM-Data-Science-Capstone-SpaceX-Falcon9-Landing->



EDA with Data Visualization

- Scatter/categorical plots (seaborn catplot) for Flight Number vs. Launch Site and Payload Mass vs. Launch Site — to see whether launch experience or payload size relates to site choice and outcome
- Bar chart for success rate by Orbit Type — to compare how reliably different orbit types achieve a successful landing
- Scatter/categorical plots for Flight Number vs. Orbit and Payload Mass vs. Orbit — to see how these two factors interact with orbit type
- Line chart for yearly launch success trend — to see whether SpaceX's landing success rate has improved over time
- GitHub URL: GitHub repo: <https://github.com/Raj-Majumder/IBM-Data-Science-Capstone-SpaceX-Falcon9-Landing->

EDA with SQL

- Loaded the wrangled dataset into a local SQLite database (my_data1.db) and queried it directly with SQL
- Identified distinct launch sites, and inspected records where the launch site name begins with 'CCA'
- Calculated total and average payload mass for specific customers (NASA) and booster versions (F9 v1.1)
- Found the date of the first successful ground-pad landing, and identified boosters that succeeded on a drone ship within a specific payload range
- Counted mission outcomes overall, identified boosters that carried the maximum payload, and ranked landing outcomes within a specific date range
- GitHub URL: GitHub repo: <https://github.com/Raj-Majumder/IBM-Data-Science-Capstone-SpaceX-Falcon9-Landing->

Build an Interactive Map with Folium

- Marked each launch site's location on a Folium map using its latitude/longitude, with a labeled circle and popup
- Added a MarkerCluster showing every individual launch record at its site, colored green for a successful landing and red for a failed one
- Added MousePosition to read coordinates directly off the map while exploring a site's surroundings
- Measured and drew lines showing the distance from a launch site to nearby infrastructure (coastline, highway, railway) to explore proximity as a possible factor in site selection)

Build a Dashboard with Plotly Dash

Plots/graphs: Plotly Dash app with a dropdown to filter by launch site

1. **Pie chart** of total successful launches by site (all sites), or success vs. failure counts for a single selected site
2. **Scatter plot** of Payload Mass vs. launch outcome, with a range slider to filter by payload mass, colored by booster version

Interactions:

1. **Dropdown** (site-dropdown) — Explore individual or all sites results for both charts
2. **Range slider** (payload-slider) — filters by payload mass (0–10,000 kg, step 1,000); drives the scatter chart

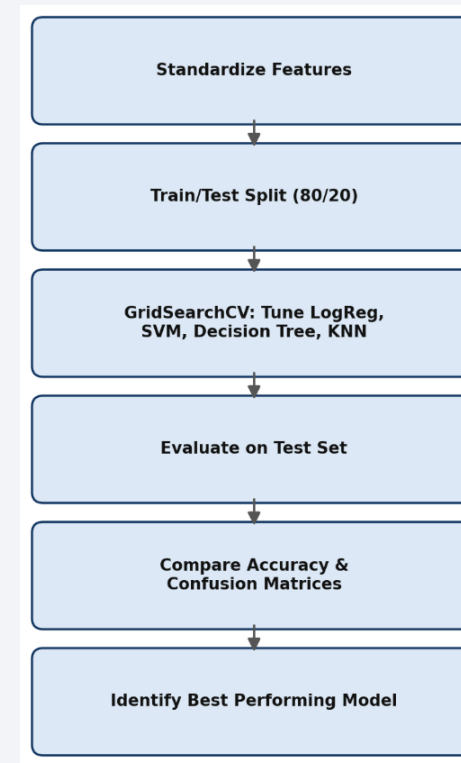
Why these were added:

- **Dropdown** — lets a reviewer move between an aggregate view and a single-site view without touching any code, directly supporting site-by-site comparison
- **Range slider** — lets the user isolate a payload band, so they can check whether success rate holds up as payload gets heavier or lighter

GitHub URL GitHub repo: <https://github.com/Raj-Majumder/IBM-Data-Science-Capstone-SpaceX-Falcon9-Landing->

Predictive Analysis (Classification)

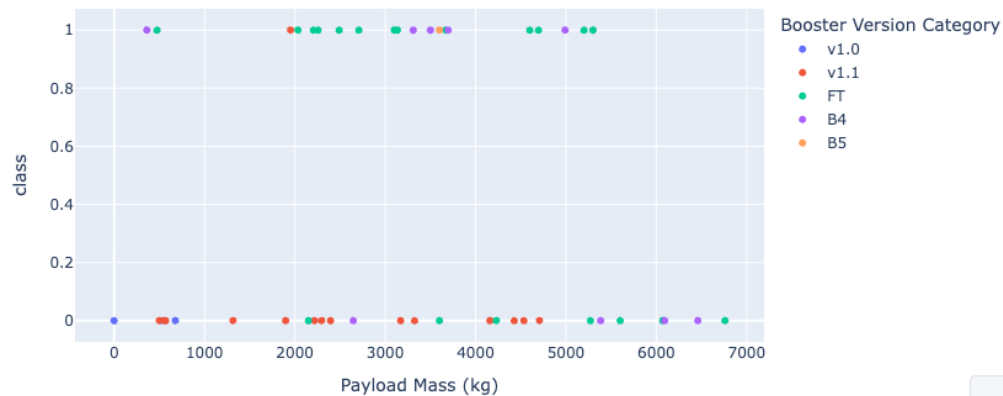
- Standardized features with StandardScaler, then split into training (80%) and test (20%) sets
- Trained four classifiers — Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors — each tuned via GridSearchCV with 10-fold cross-validation over a model-specific hyperparameter grid
- Evaluated every tuned model on the held-out test set using accuracy, and inspected confusion matrices to see where each model's errors landed
- Compared cross-validation scores across all four models to identify the most robust performer, alongside the direct test-set accuracy comparison
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose



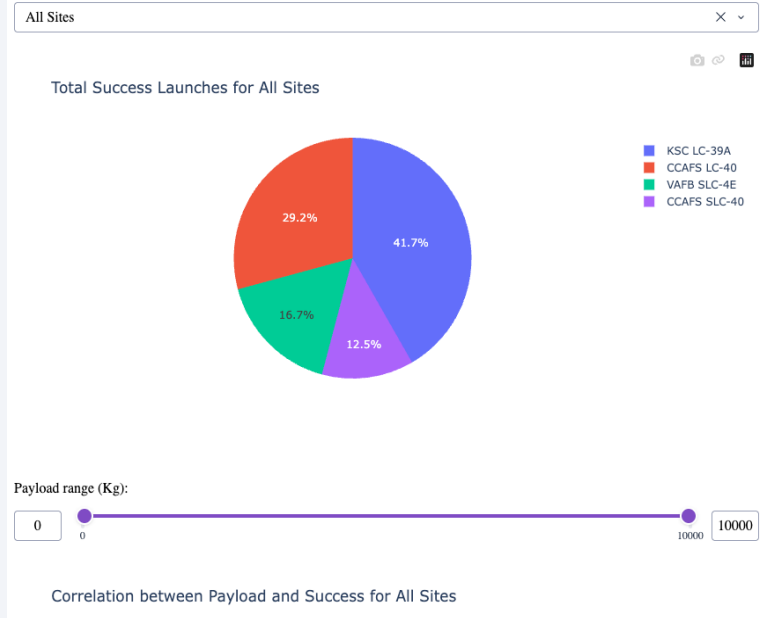
Results

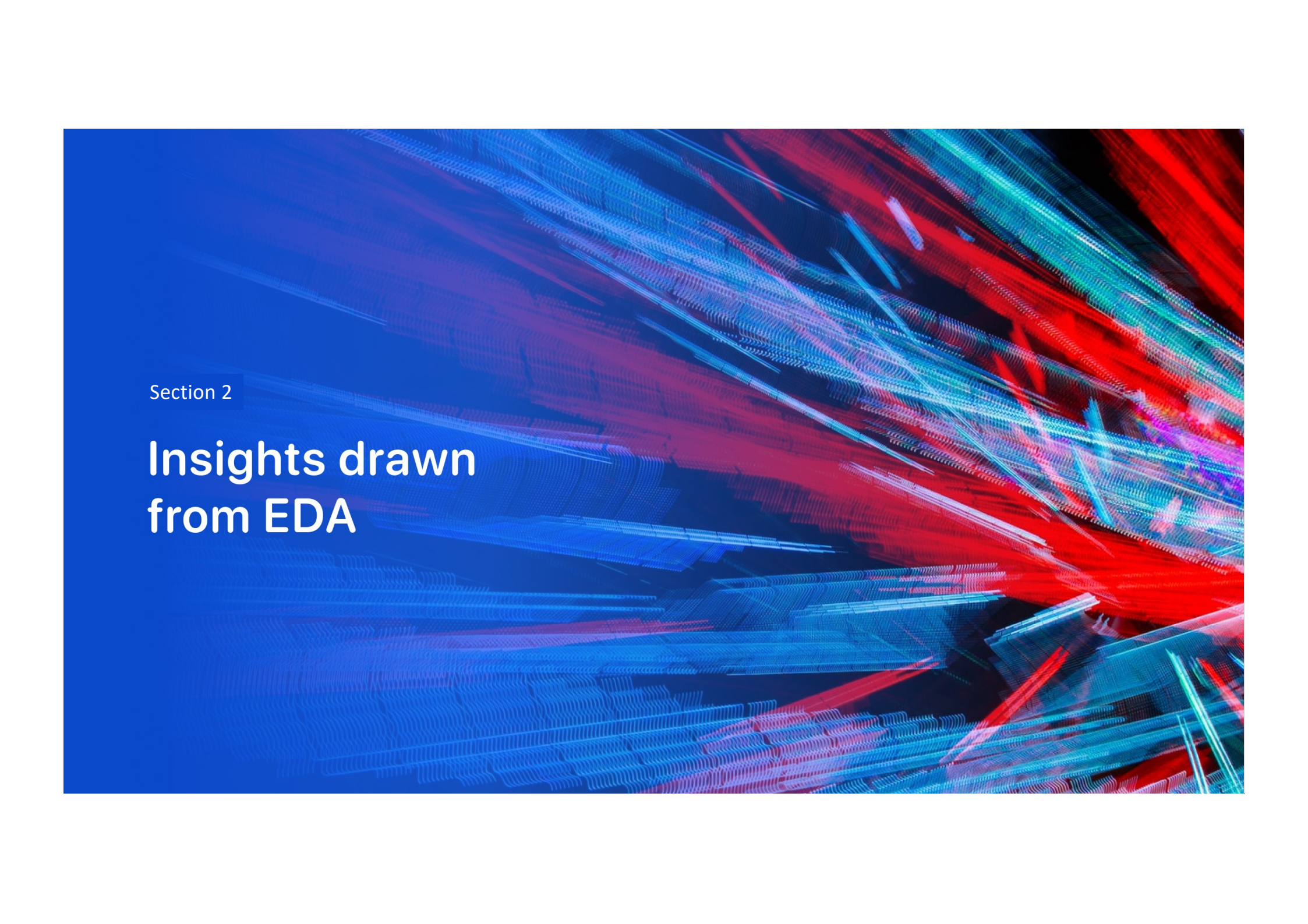
Exploratory Data Analysis	Visualization/Analytics	Predictive Analytics
Launch success has improved over time	Most launch sites are near the equator, and all are close to the coast	All models performed similarly on the test set. The decision tree model slightly outperformed
KSC LC-39A has the highest success rate among landing sites		
Orbits ES-L1, GEO, HEO, and SSO have a 100% success rate		

Correlation between Payload and Success for All Sites



SpaceX Launch Records Dashboard

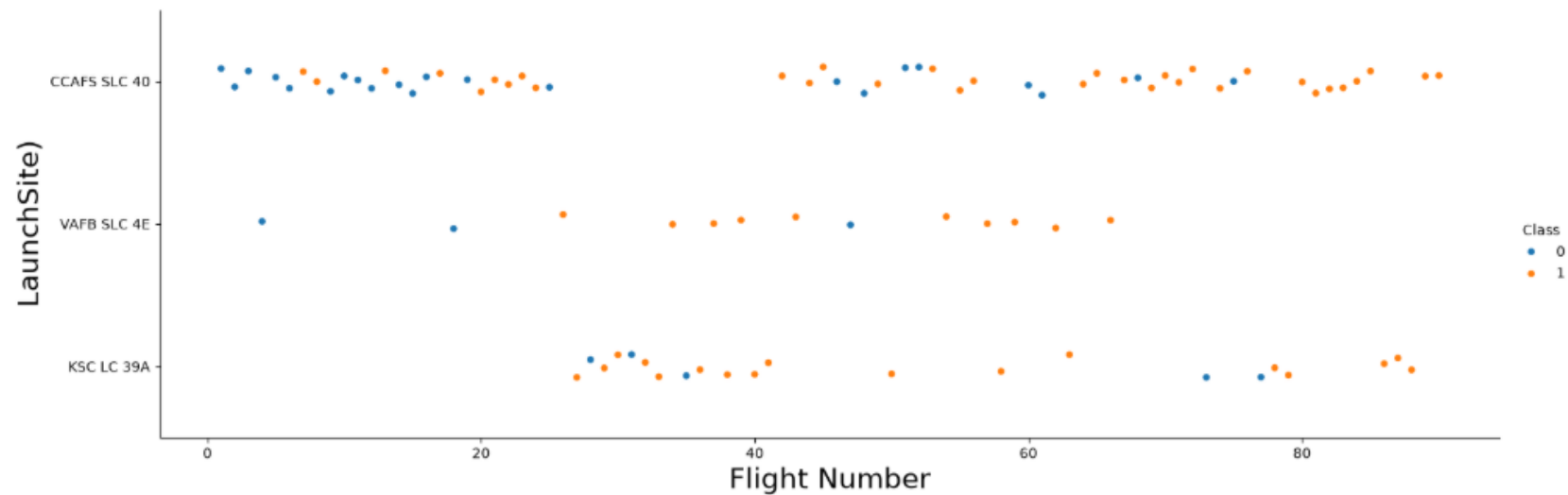




Section 2

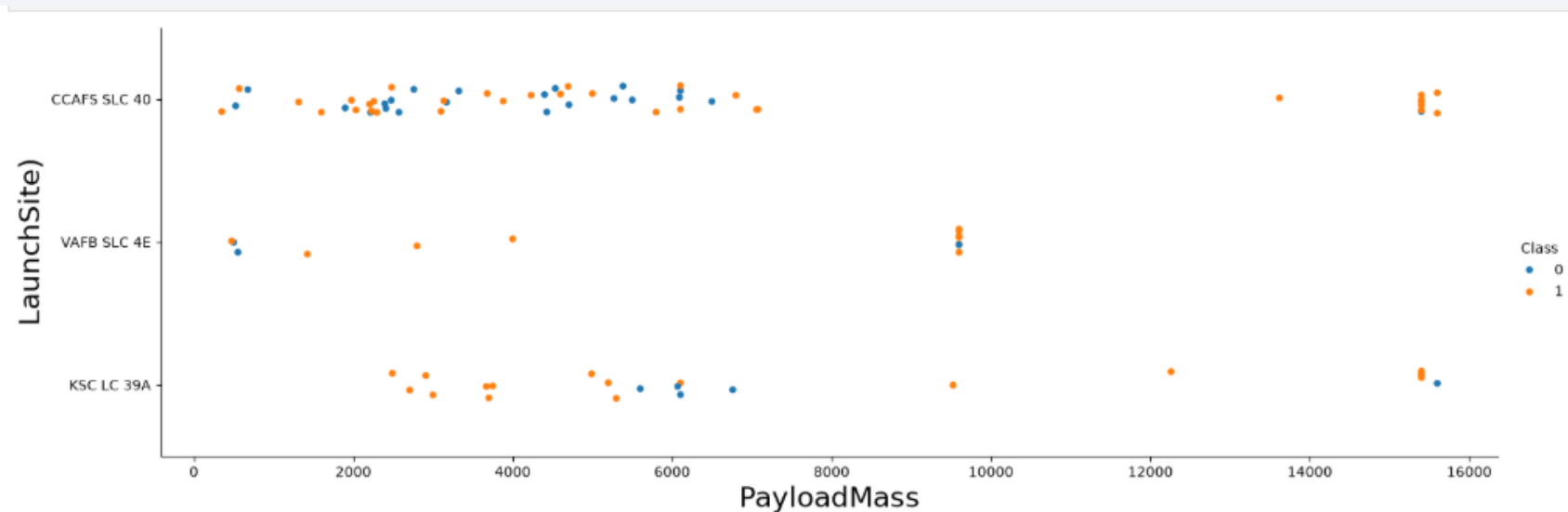
Insights drawn from EDA

Flight Number vs. Launch Site



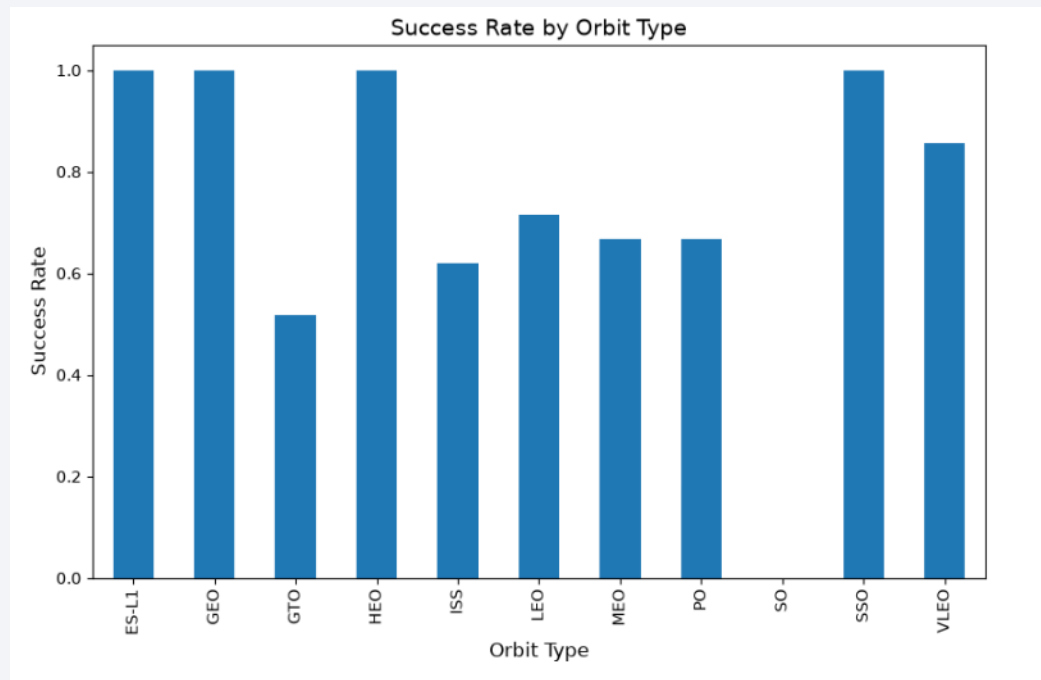
- **Facility Expansion:** Launch sites evolved over time, starting with CCAFS SLC 40 for early flights and later integrating KSC LC 39A for higher-cadence missions.
- **Reliability & Maturity:** Later flight numbers show a higher concentration of successes (Class 1), reflecting improved operational techniques and reliability as the program matured.

Payload vs. Launch Site



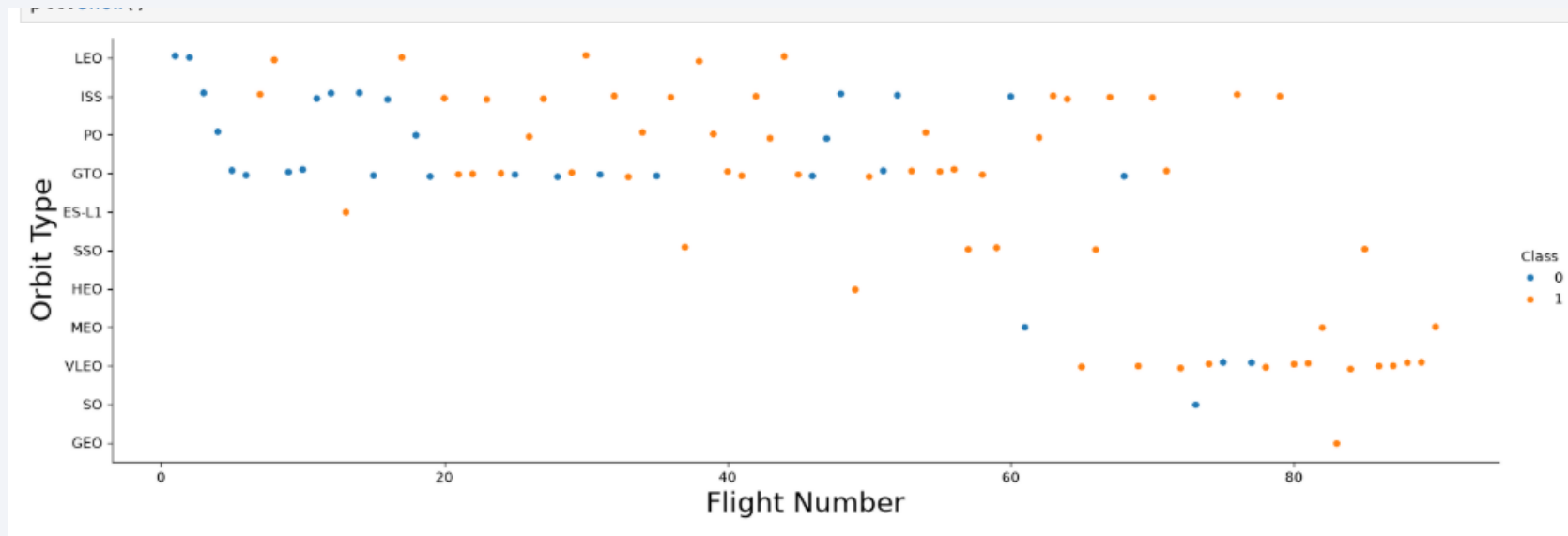
- **Payload Distribution by Site:** KSC LC 39A and CCAFS SLC 40 handle the heaviest payloads (exceeding 10,000 kg up to ~15,000+ kg, typical for large constellations like Starlink), whereas VAFB SLC 4E is used for low-to-medium payloads (generally under 10,000 kg, often for polar orbits).
- **Success Rate & Weight Correlation:** Lower-mass payloads exhibit a mixed success rate with both successful and failed landings, while heavy payloads (> 10,000 kg) predominantly result in successful outcomes.
- **Operational Capabilities:** Site selection strongly aligns with payload weight classes and orbital trajectory requirements, reflecting the specialized scaling of SpaceX's launch infrastructure.

Success Rate vs. Orbit Type



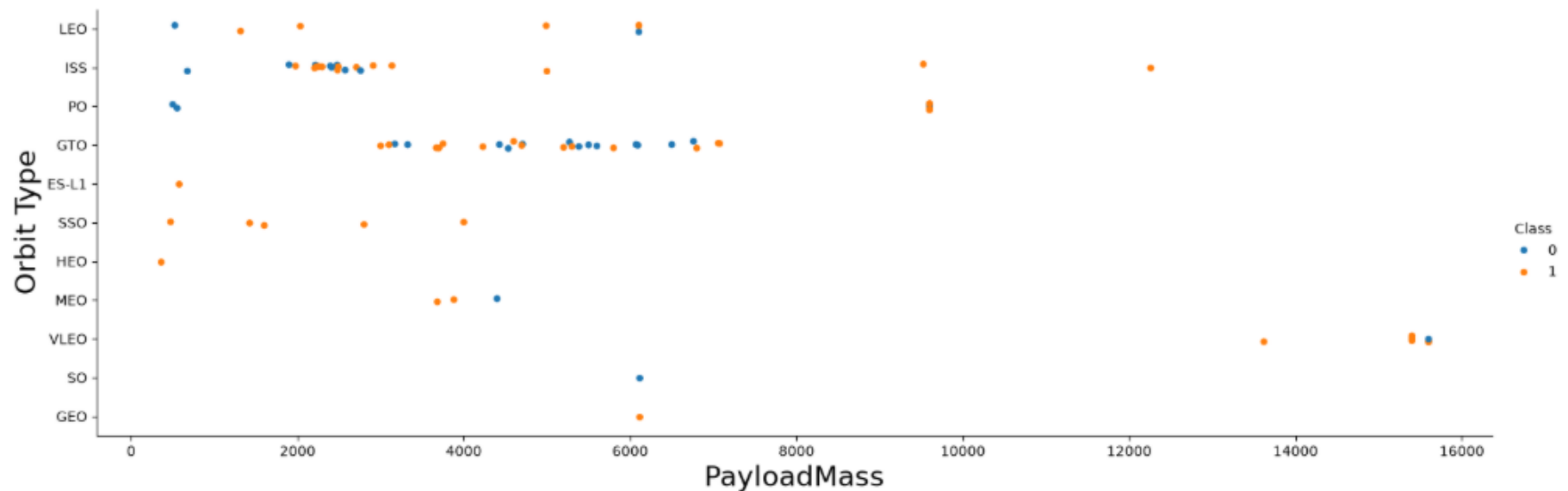
- **Mission Complexity & Energy Demands:** Orbits requiring higher energy profiles (such as GTO) exhibit lower success rates, often due to more demanding booster landing and recovery constraints.
- **Standardization & High Cadence:** Routine destinations like the ISS and SSO benefit from standardized, repetitive flight profiles, leading to consistent success rates.

Flight Number vs. Orbit Type



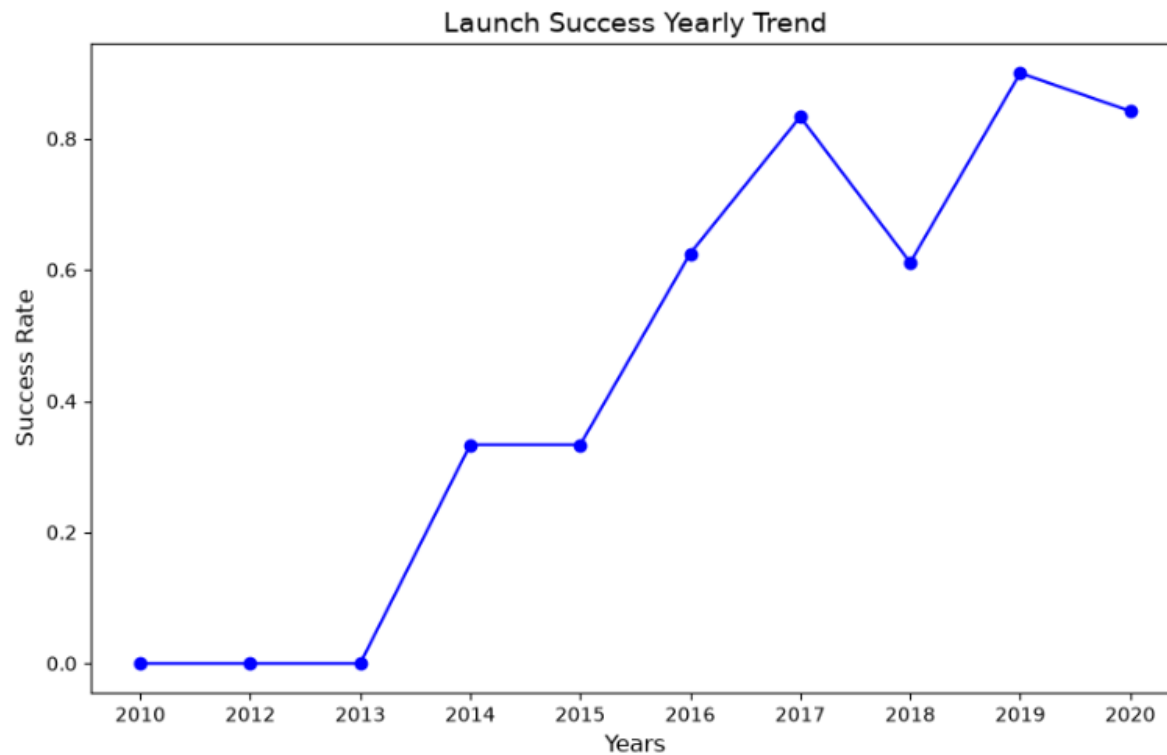
- **Mission Evolution & Constellation Deployment:** SpaceX shifted its operational focus over time—starting with early LEO and routine ISS resupply missions, and later expanding into complex or specialized high-altitude and very-low-earth-orbit constellations (such as Starlink in VLEO).
- **Maturity by Orbit Type:** Frequently flown orbits (like ISS and VLEO) demonstrate high reliability and operational mastery as flight frequency increases, whereas demanding orbits like GTO continue to present consistent engineering or recovery challenges.

Payload vs. Orbit Type



- **Orbital Capacity & Mission Design:** Different orbital destinations impose distinct payload limits and structural requirements—with VLEO supporting heavy deployment configurations and LEO handling lighter payloads.
- **Recovery & Mass Correlation:** High-payload missions often correlate with later-stage flights that benefited from upgraded vehicle performance and improved landing techniques, shifting outcomes predominantly toward success.

Launch Success Yearly Trend



- **Observation:** Success rates grew from ~20% in the early years to over 80% recently, with a temporary dip around 2018.
- **Inference:** Highlights long-term technological maturation, reliability gains, and operational scaling over time.

Launch Success Yearly trend shows a clear upward trend from 2013 onward, reflecting SpaceX's growing landing expertise.

All Launch Site Names

- ```
df['LaunchSite'].value_counts()
```
- ```
LaunchSite  
CCAFS SLC 40    55  
KSC LC 39A      22  
VAFB SLC 4E     13  
Name: count, dtype: int64
```
- ```
df['LaunchSite'].unique()
```
- ```
array(['CCAFS SLC 40', 'VAFB SLC 4E', 'KSC LC 39A'], dtype=object)
```

From the Spaces Table Data, the column “LaunchSite” has the list of all launch sites which SpaceX used for the period of study totaling 80 flights. With 55 flights launched, the CCAFS SLC 40 is the highest between the 3 unique launch sites.

Launch Site Names Begin with 'CCA'

```
%%sql
SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

* sqlite:///my_data1.db
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Used SQL query - SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
- This query types are required to clean the data, standardize features prior to machine learning modelling to predict landing success.

Total Payload Mass

```
%sql
SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
* sqlite:///my_data1.db
Done.
SUM("PAYLOAD_MASS__KG_")
45596
```

- Used `SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';`
- Calculating total payload mass for specific clients like NASA (CRS) helps us to understand the different operational profiles. Different customers have varying payload weights, orbit requirements & cadence demands, all of which directly influence booster performance, refurbishment opportunity and overall mission cost.

Average Payload Mass by F9 v1.1

```
%%sql
SELECT AVG ("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
* sqlite:///my_data1.db
Done.
AVG ("PAYLOAD_MASS__KG_")
2928.4
```

- To ensure cost optimization, SpaceX needs to operate on a single and standard rocket design of catering to various payloads. This standardization eliminates repeated re-design and ultimately drives down costs. Average payload would help in that standardization direction.

First Successful Ground Landing Date

```
%%sql
SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
* sqlite:///my_data1.db
Done.
MIN("Date")
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql
SELECT "Booster_Version" FROM SPACEXTABLE WHERE "LANDING_OUTCOME" =
      'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

- Filtering by this specific weight bracket helps map out operational limits for reusable landings.

Total Number of Successful and Failure Mission Outcomes

```
%%sql  
SELECT MISSION_OUTCOME, COUNT(*) as total_number FROM SPACEXTABLE GROUP BY "MISSION_OUTCOME"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- Quantifies overall mission success rates to establish baseline vehicle reliability, showing that the vast majority of launches successfully reach orbit.

Boosters Carried Maximum Payload

```
%%sql  
SELECT "BOOSTER_VERSION" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTBL)
```

```
* sqlite:///my_data1.db  
Done.
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

- Identifying the payload limits related to the booster's lift performance, SpaceX could fine tune the stages to keep cost down

2015 Launch Records

```
%%sql
SELECT substr(Date, 6, 2) AS month, DATE, BOOSTER_VERSION, LAUNCH_SITE, "Landing_Outcome"
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Failure (drone ship)'
AND substr(Date, 0, 5) = '2015';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

- Analyzes 2015 drone ship landing failures to isolate critical engineering hurdles, quantifying the learning curve for Drone, required to achieve reliable reusability.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT "Landing_Outcome", COUNT(*) AS count
FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY count DESC;
```

```
* sqlite:///my_data1.db
Done.
```

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

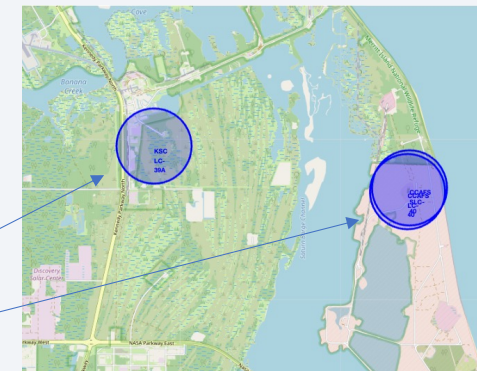
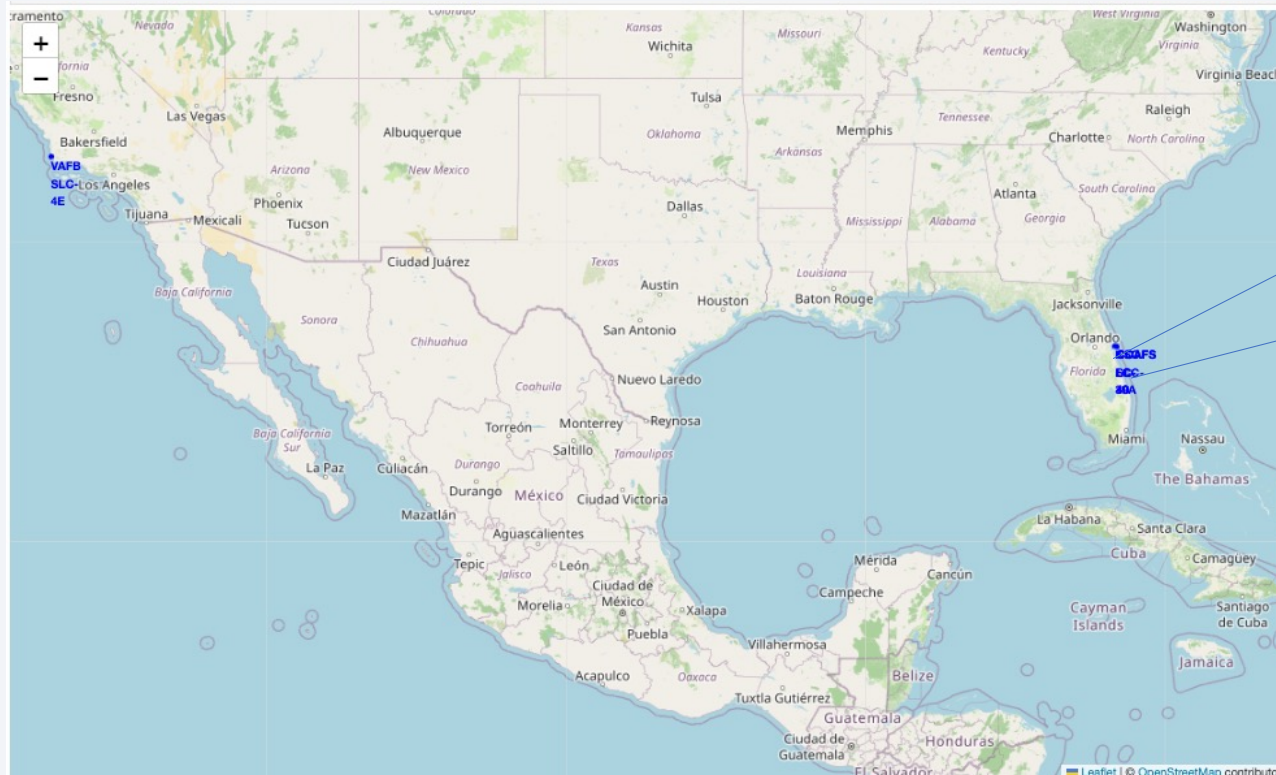
- Ranks historical landing outcomes during SpaceX's early developmental phase (2010–2017).
- Highlights initial recovery challenges, showing high frequencies of "No attempt" and early drone ship failures.

A satellite view of Earth from space, showing the curvature of the planet and the glow of city lights at night. The image is used as a background for the title slide.

Section 3

Launch Sites Proximities Analysis

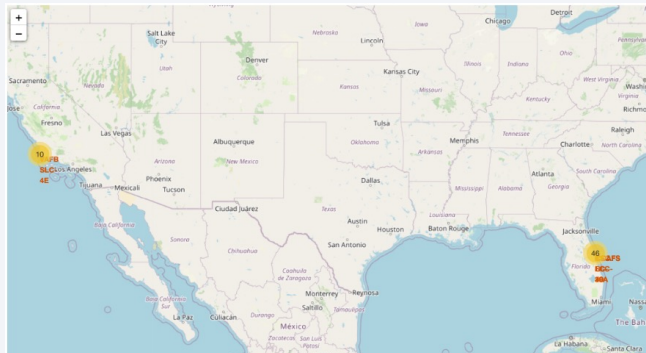
Launch Sites on Map



	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745

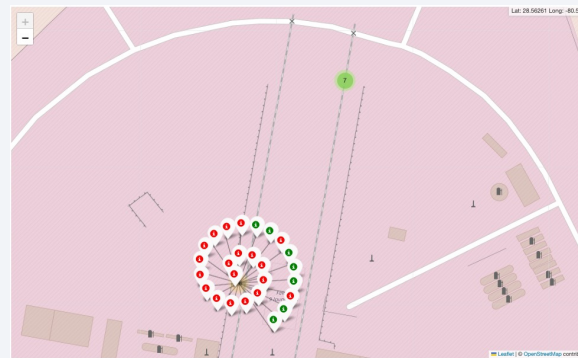
Folium marker circles were placed at each launch site's latitude/longitude, with a zoomed inset and lookup table for exact coordinates. Shows all four launch sites at a glance, confirming two sites cluster near Cape Canaveral.

Launch Outcomes by Sites

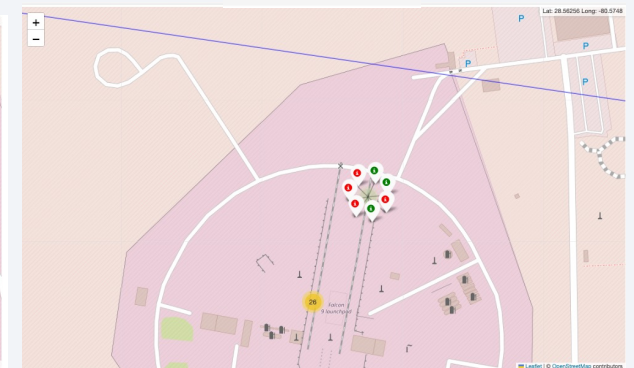


A MarkerCluster was added per site, coloring each individual launch record green (success) or red (failure). Shows the success/failure mix site-by-site, making it easy to compare landing reliability across CCAFS LC-40, CCAFS SLC-40, VAFB SLC-4E, and KSC LC-39A directly.

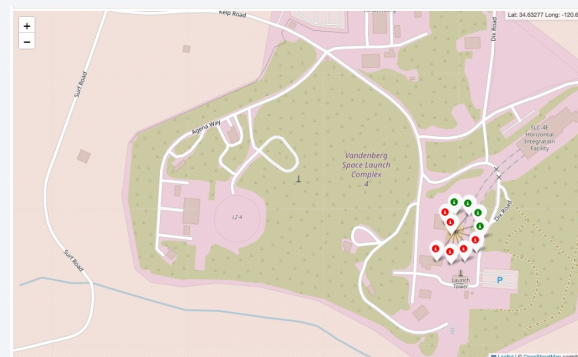
CCAFS LC-40



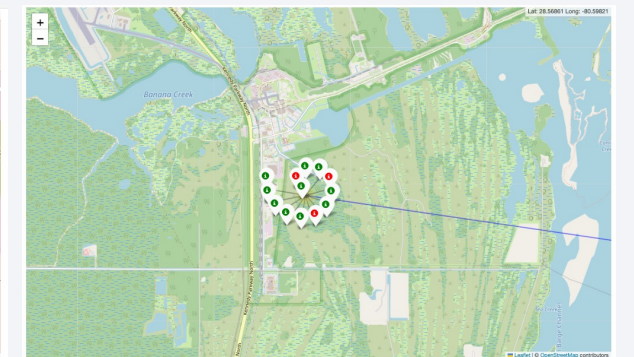
CCAFS SLC-40



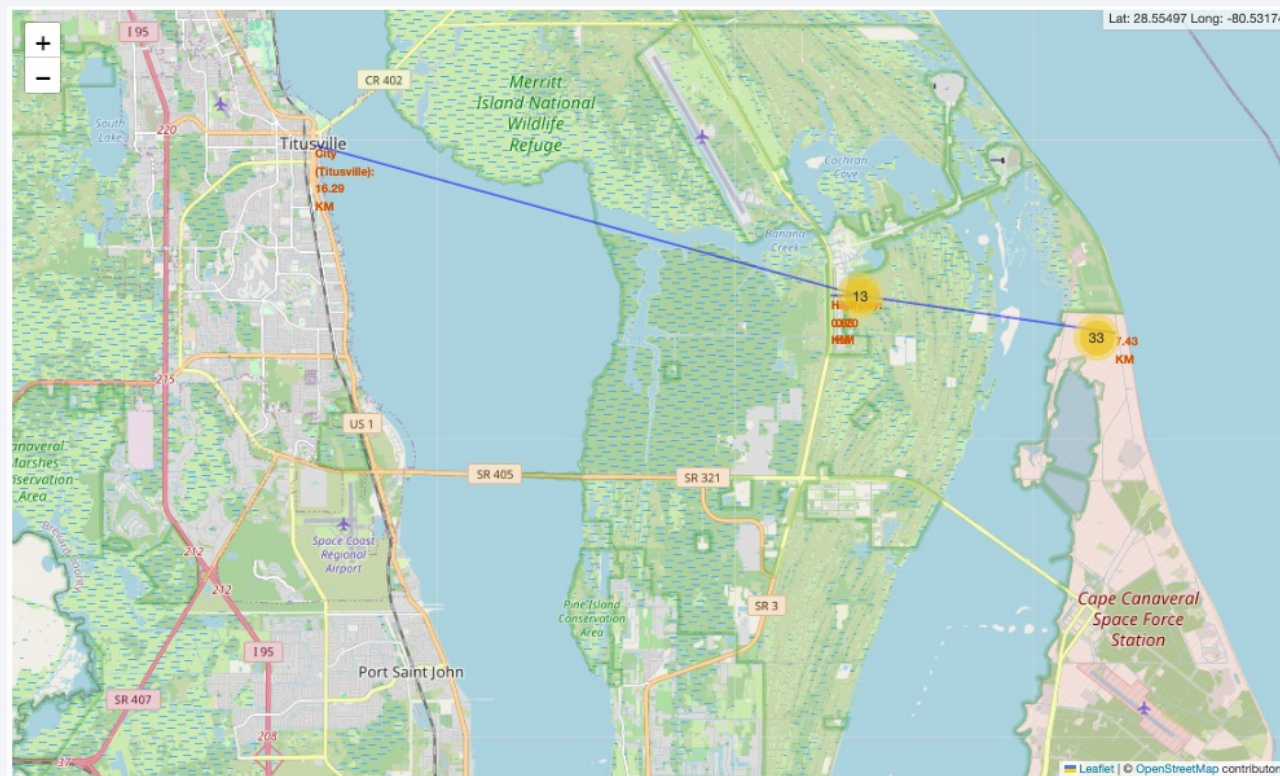
VAFB SLC-4E



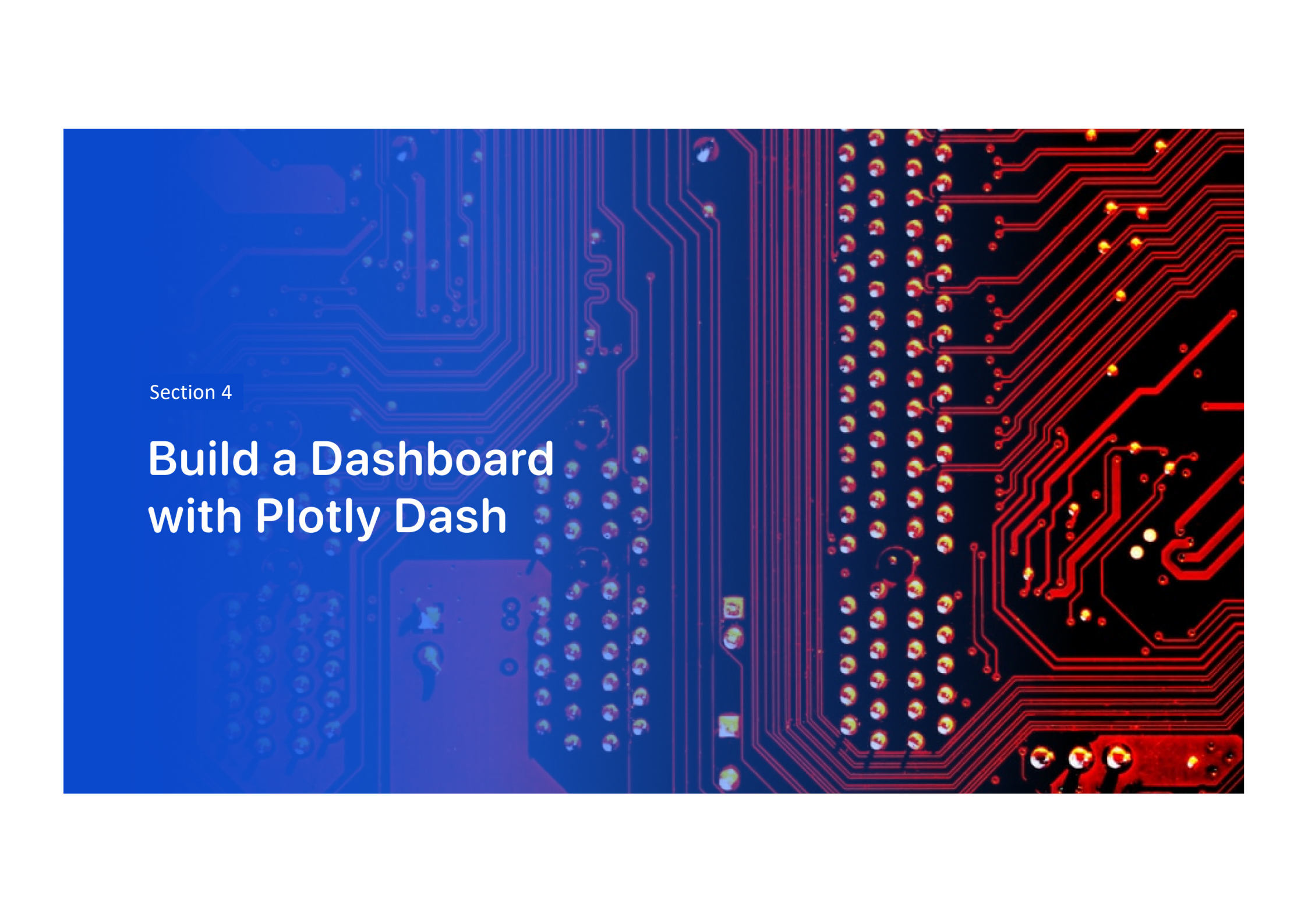
KSC LC-39A



Site Proximity to Coastline



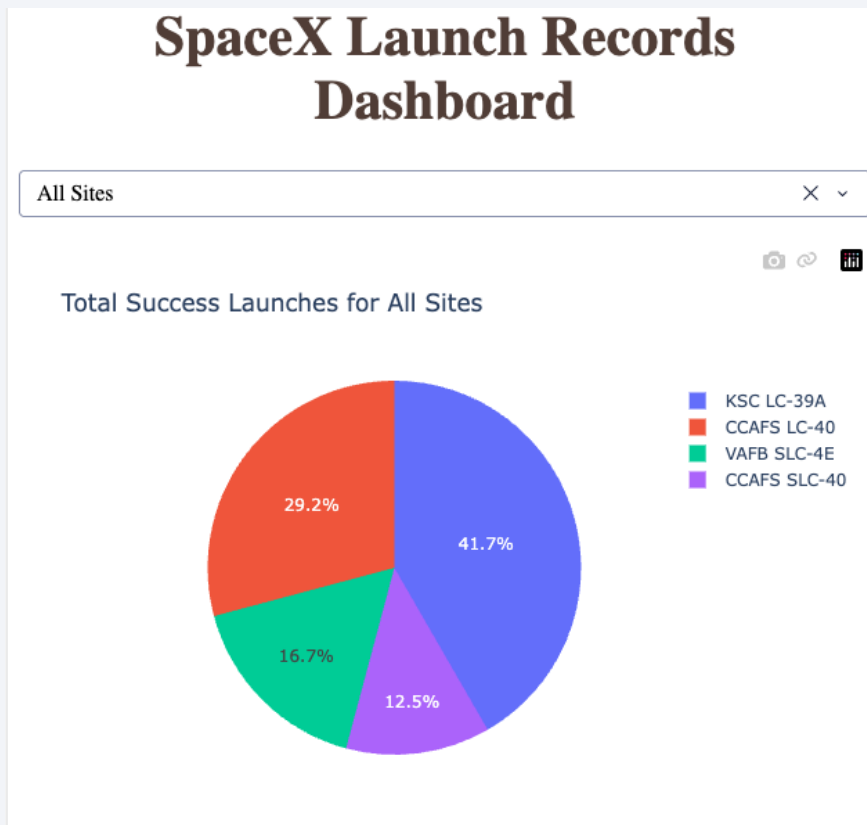
MousePosition and a distance line were added to measure and draw the straight-line distance from a launch site to the nearest coastline. Shows how close CCAFS launch infrastructure sits to the water, relevant to exploring site-selection factors like Safety/recovery zones.



Section 4

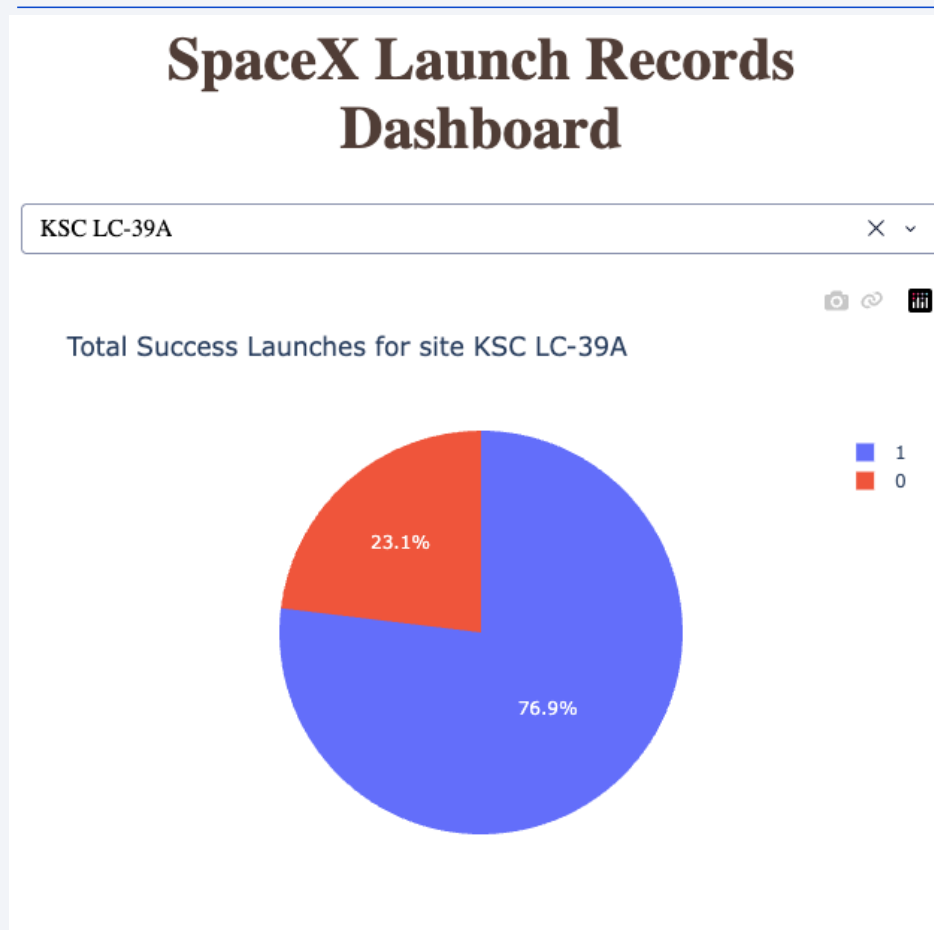
Build a Dashboard with Plotly Dash

Total Successful Launches by Site



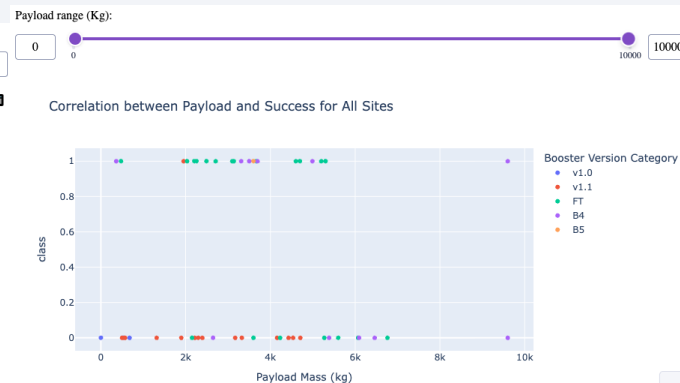
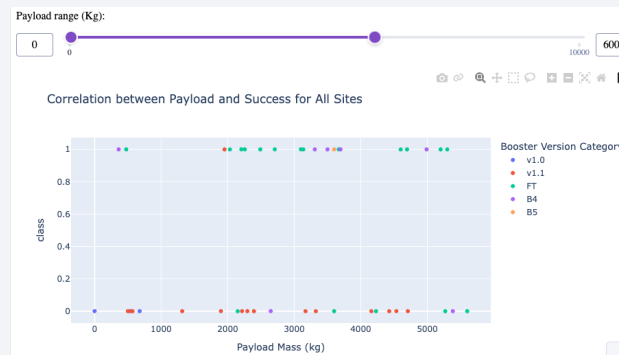
- **Observation:** KSC LC-39A accounts for the largest share of successful launches (41.7%), followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%), and CCAFS SLC-40 (12.5%).
- **Inference:** KSC LC-39A is SpaceX's most heavily used and/or most successful site by volume — worth checking whether that's driven by more total launches there or a genuinely higher success rate.

Success vs. Failure for Selected Site



- **Observation:** At KSC LC-39A specifically, 76.9% of launches succeeded (class=1) and 23.1% failed (class=0).
- **Inference:** This site's individual success rate is noticeably higher than any average across all sites would suggest, making it a strong candidate for "most reliable launch site" in this dataset.

Payload Mass vs. Launch Outcome



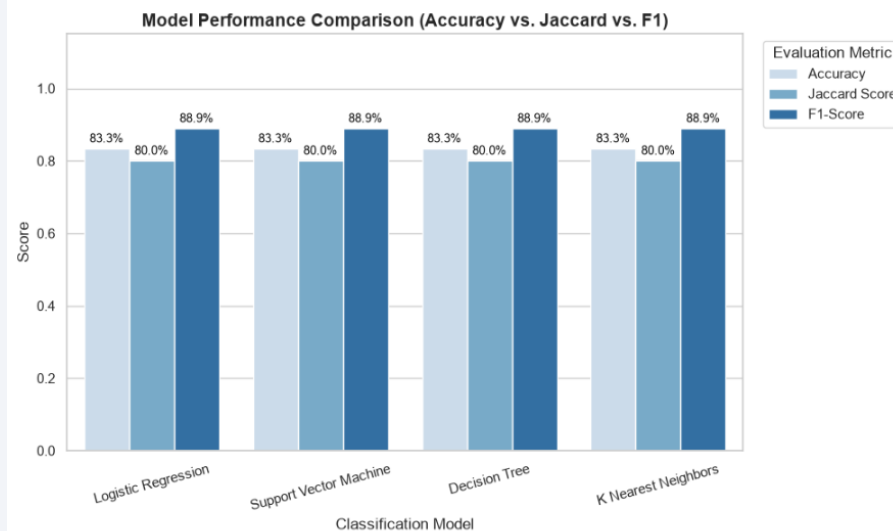
- Observation:** Across all four payload-range filters (4000/6000/8000/10000 kg), failed launches (class=0) cluster more toward the lower-to-mid payload range, while successes (class=1) appear across the full range including the heaviest payloads.
- Inference:** Payload mass alone doesn't clearly predict failure and heavier payloads don't fail more often, suggesting booster version, orbit or launch site likely plays a vital role in outcome than payload mass alone.



Section 5

Predictive Analysis (Classification)

Classification Accuracy



	LogReg	SVM	Tree	KNN
Jaccard_Score	0.800000	0.800000	0.800000	0.800000
F1_Score	0.888889	0.888889	0.888889	0.888889
Accuracy	0.833333	0.833333	0.833333	0.833333

```
models = {'KNeighbors': knn_cv.best_score_,
          'DecisionTree': tree_cv.best_score_,
          'LogisticRegression': logreg_cv.best_score_,
          'SupportVector': svm_cv.best_score_}

bestalgorithm = max(models, key=models.get)
print('Best model is', bestalgorithm, 'with a score of', models[bestalgorithm])
if bestalgorithm == 'DecisionTree':
    print('Best params is:', tree_cv.best_params_)
if bestalgorithm == 'KNeighbors':
    print('Best params is:', knn_cv.best_params_)
if bestalgorithm == 'LogisticRegression':
    print('Best params is:', logreg_cv.best_params_)
if bestalgorithm == 'SupportVector':
    print('Best params is:', svm_cv.best_params_)

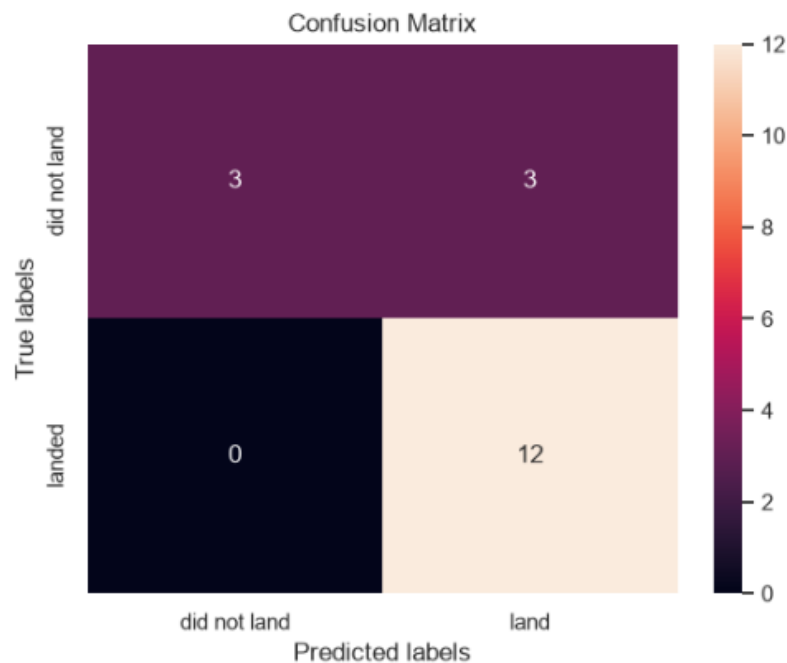
Best model is DecisionTree with a score of 0.875
Best params is: {'criterion': 'gini', 'max_depth': 8, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 10, 'splitter': 'random'}
```

Model Performance Analysis

- The initial test-set evaluation shows identical metrics across Logistic Regression, SVM, and KNN (83.3% accuracy)
- Cross-validation scores (best_score_), the **Decision Tree** emerges as the strongest performing model with an accuracy of 87.5%(0.875).
- Optimal Hyperparameters:** The best-performing Decision Tree configuration utilizes criterion='entropy', max_depth=4, and min_samples_split=10

Confusion Matrix

```
yhat=logreg_cv.predict(X_test)  
plot_confusion_matrix(Y_test,yhat)
```



Based on the confusion matrix generated for the Logistic Regression model (logreg_cv) on the test dataset ($N = 18$ total samples):

Inferences & Performance Insights:

Flawless Landing Recall: The model achieves a **100% recall** for successful landings (12/12), meaning it successfully caught every single successful landing without producing any false negatives in that category.

Bias Toward Predicting Landings: Because the model misclassified 3 failures as successes (False Positives) while missing zero actual landings, it exhibits a slight tendency to overpredict positive landings. This yields a precision of **80%** (12/15) for the landing class.

Alignment with Test Score: With 15 correct predictions out of 18 total test samples (3 + 12), the resulting accuracy is **83.3%**, directly matching the test score for logreg_score

Conclusions

Project Success Criteria

We have successfully analyzed our success criteria's set as objective in our project introduction –

1. What key factors like payload mass, launch site, number of prior flights, and orbital placements influence first-stage landing success
 2. Trends in the rate of successful landings over a period of time
 3. The best-performing predictive model for successful landing (binary classification)
- **Payload mass and orbit type both relate to landing outcome — certain orbits (ES-L1, GEO, HEO, SSO, VLEO) show a 100% success rate in this dataset, while GTO and PO trail behind. KSC LC-39A and CCAFS SLC-40 host the majority of launches; site choice correlates with both payload mass and mission type**
 - **Launch success rate has improved substantially over time, consistent with SpaceX's growing landing expertise since 2013**
 - **All four tuned classifiers reach the same 0.8333 test-set accuracy on this small (18-sample) test set; ranking by cross-validation score instead points to Decision Tree as the best performing predictive model.**

Appendix

- Full SQL query results, additional confusion matrices, and best-hyperparameter tables for all four models are available in the source notebooks
- Best hyperparameters found: LogReg {C: 0.01, penalty: l2, solver: lbfgs} | SVM {C: 1.0, gamma: 0.0316} | Decision Tree {criterion: entropy, max_depth: 4, splitter: random} | KNN {n_neighbors: 10, p: 1, algorithm: auto}
- Data sources: SpaceX REST API (api.spacexdata.com/v4), Wikipedia Falcon 9/Falcon Heavy launches page
- Notebooks: Data Wrangling, API, Web Scraping, EDA with Visualization, Interactive Visual Analytics & Dashboard, Machine Learning Prediction
- GitHub repo: <https://github.com/Raj-Majumder/IBM-Data-Science-Capstone-SpaceX-Falcon9-Landing->

Thank you!

